CSA0404 – OPERATING SYSTEMS

EXPERIMENT 31

QUESTION

Construct a C program to simulate the First in First Out paging technique of memory management.

PROGRAM

```c
#include <stdio.h>

#define MAX_FRAMES 3

void printFrames(int frames[], int n) {
    for (int i = 0; i < n; i++) {
        if (frames[i] == -1) {
            printf(" - ");
        } else {
            printf(" %d ", frames[i]);
        }
    }
    printf("\n");
}

int main() {
    int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3,
    int n = sizeof(referenceString) / sizeof(referenceStri
    int frames[MAX_FRAMES];
    int framePointer = 0;

    for (int i = 0; i < MAX_FRAMES; i++) {
        frames[i] = -1;
    }

    printf("Reference String: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", referenceString[i]);
    }
    printf("\n\n");

    printf("Page Replacement Order:\n");
```

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2

Page Replacement Order:
Page 7 ->  7  -  -
Page 0 ->  7  0  -
Page 1 ->  7  0  1
Page 2 ->  2  0  1
Page 3 ->  2  3  1
Page 0 ->  2  3  0
Page 4 ->  4  3  0
Page 2 ->  4  2  0
Page 3 ->  4  2  3
Page 0 ->  0  2  3


...Program finished with exit code 0
Press ENTER to exit console.
```